

Simplifying User Interactions

A Large Action Model for web Automation

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Explanation & goals

Manually executing repetitive YouTube actions is time-consuming. Our solution automates this process, reducing human intervention by leveraging AI and vision-based verification.

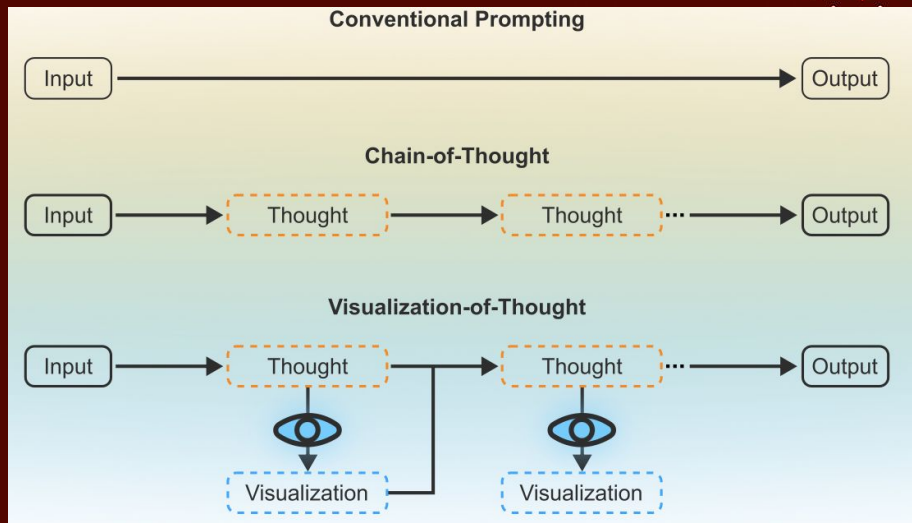
Explanation

- Automating Youtube tasks using LLMs and Selenium
- Translates natural language instructions into executable actions
- visual verification using CLIP, allowing it to analyze screenshots and ensure steps are executed correctly

Goals

- To develop framework for automating YouTube tasks
- To Improve reliability by using visual feedback to verify whether steps were executed as intended.
- To minimise failures and unnecessary retries while maintaining high accuracy in task completion.

Prior works



- Introduced by researchers from Google DeepMind
- Our system adapts this concept by integrating CLIP-based vision analysis to validate step execution dynamically.

Prior works



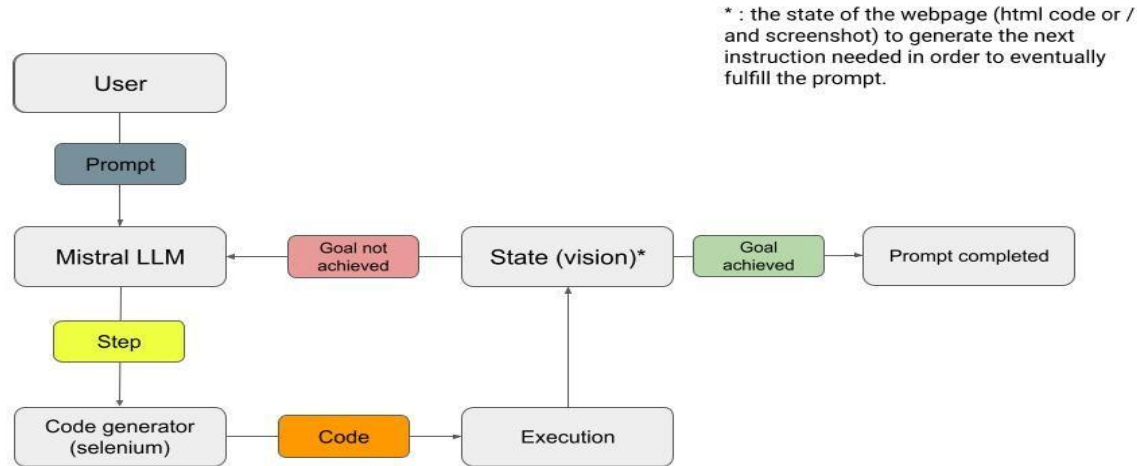
- Pywinassistant
- A framework Computer-Using-Agent that fully operates graphical-user-interfaces (GUIs) by using only natural language.

Prior works

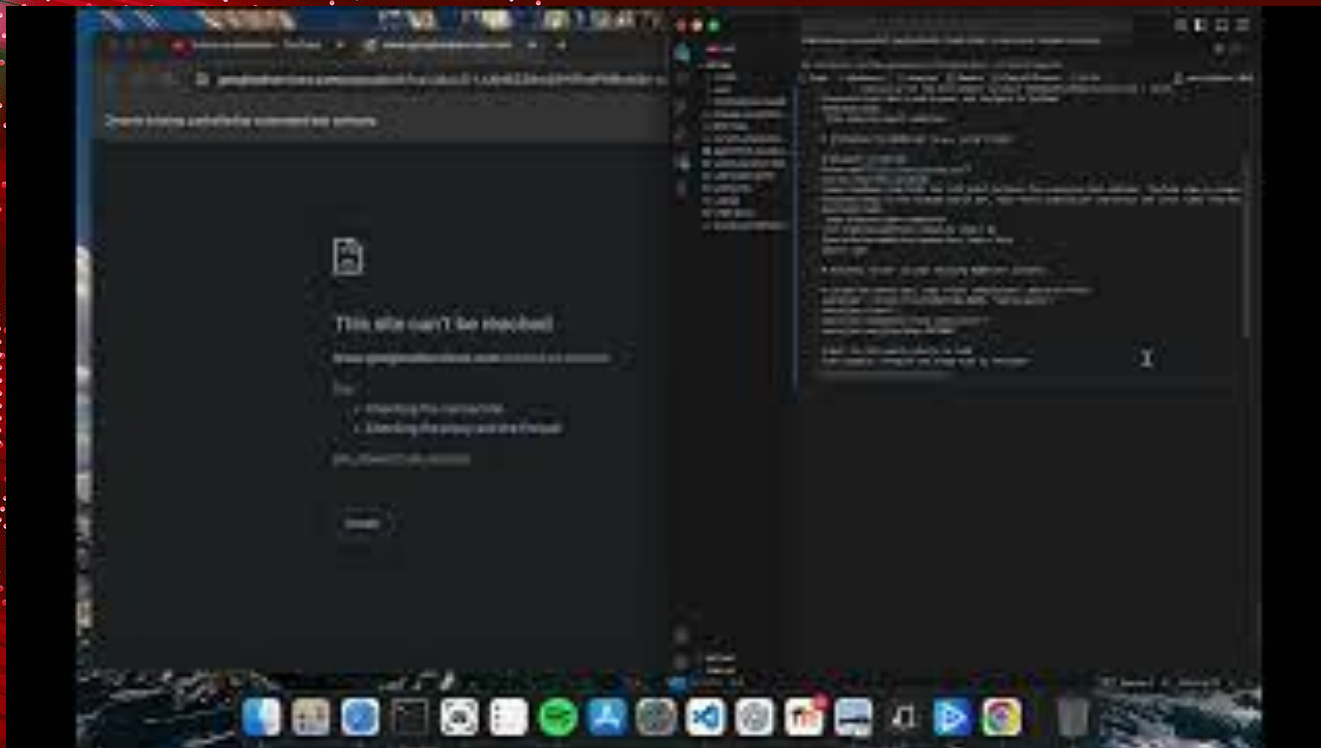


- OpenAI Operator: Cloud-Based Task Execution
- Operator can “see” (through screenshots) and “interact” (using all the actions a mouse and keyboard allow) with a browser, enabling it to take action on the web without requiring custom API integrations.

Systeme architecture



Demo



Experiments

- 10 tasks with 4 trails each
- 60 % success rate
- With an average time of 30 seconds for the successful ones

Limitations

API costs & Rate limits

Using Hugging Face APIs and OpenAI models incurs costs and API rate limits, which may impact scalability.

Execution time

The iterative nature of the automation can make it slower than human navigation

UI changes

Visual elements and their names can change which affect reliability

Achievements

✓ Implemented Selenium-based automation framework

✓ Integrated Mistral LLM for step generation

✓ Developed dynamic code generation from LLM instructions

✓ Integrated CLIP-based vision model to help guide the system

✓ Enhanced task execution loop with user feedback handling



Future work & Conclusion

- 🚀 **Task Completion Detection** : Automate end-of-task recognition to eliminate manual confirmation
- 🚀 **Improved Code Generation** : Explore DeepSeek R1 for more reliable and cheaper Selenium script generation
- 🚀 **Enhanced Visual Feedback** : Integrate OCR-based text detection alongside CLIP for better web state understanding
- 🚀 **Generalized Web Automation** : Expand beyond YouTube to other websites while minimizing prompt engineering

THANK YOU FOR LISTENING!

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